CBSE Test Paper 02

Chapter 3 Motion in A Straight Line

1. A particle moves along the x axis. Its position is given by the equation x = 2.00 + 3.00t - $4.00t^2$ with x in meters and t in seconds. Determine its velocity in m/s when it returns to the position it had at t = 0. **1**
   a. -2.54
   b. -3.0
   c. -2.75
   d. -4.02
2. If the position- time graph is a straight line parallel to the time axis **1**
   a. The velocity is decreasing
   b. The velocity is constant but non zero
   c. The velocity is zero
   d. The velocity is increasing
3. A truck on a straight road starts from rest, accelerating at 2.00 $\mathrm{m/s^2}$ until it reaches a speed of 20.0 m/s. Then the truck travels for 20.0 s at constant speed until the brakes are applied, stopping the truck in a uniform manner in an additional 5.00 s. What is the average velocity in m/s of the truck for the motion described? **1**
   a. 15.7
   b. 16.2
   c. 154
   d. 17.5
4. A jet lands on an aircraft carrier at 30 m/s. What is its acceleration if it stops in 2.0 s? **1**
   a. 20 ms^{-2}
   b. -20 ms^{-2}
   c. -15 ms^{-2}
   d. -10 ms^{-2}
5. A truck on a straight road starts from rest, accelerating at 2.00 $\mathrm{m/s^2}$ until it reaches a speed of 20.0 m/s. Then the truck travels for 20.0 s at constant speed until the brakes are applied, stopping the truck in a uniform manner in an additional 5.00 s. How long in seconds is the truck in motion? **1**

a. 23.0
b. 37.3
c. 35.0
d. 32.0

6. Draw the position-time graph for a body at rest. **1**
7. A train is moving on a straight track with acceleration a. A passenger drops a stone. What is the acceleration of stone with respect to passenger? **1**
8. With zero speed a particle may not have non-zero velocity. Explain. **1**
9. Write the characteristics of displacement. **2**
10. Draw the displacement time graph for a uniformly accelerated motion. **2**
11. What is uniform motion? Give examples. **2**
12. Establish the kinematic equation $v^2 - u^2 = 2as$ from velocity-time graph for a uniformly accelerated motion. **3**
13. Obtain equation of motion for constant acceleration using the method of calculus. **3**
14. A jet plane beginning its take-off moves down the runway at a constant acceleration of $4.00\ m/s^2$. **3**
    i. Find the position and velocity of the plane 5.00 s after it begins to move.
    ii. If the speed of 70.0 m/s is required for the plane to leave the ground, how long a runway is required?
15. Two stones are thrown up simultaneously from the edge of a cliff 200 m high with initial speeds of 15 m/s and 30 m/s. Verify that the graph shown in Figure correctly represents the time variation of the relative position of the second stone with respect to the first. Neglect air resistance and assume that the stones do not rebound after hitting the ground. Take $g = 10\ m/s^2$. Give the equations for the linear and curved parts of the plot. **5**